Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 207 (2022) 3443–3452

Procedia Computer Science

www.elsevier.com/locate/procedia

26th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2022)

Clustering of scientific articles using natural language processing

Barbara Probierz[a,*], Jan Kozak[a], Anita Hrabia[a]

[a]*Department of Machine Learning, University of Economics in Katowice, 1 Maja, 40-287 Katowice, Poland*

Abstract

With the development of the Internet, the number of scientific articles published online has also increased significantly. In recent years there has been a steady increase in the number of articles available online. Despite providing keywords and preparing abstracts, it is often a problem for articles to be properly matched during editorial process - to match the article to the competence of a particular editor. It is also a hassle for researchers who want to be kept informed about the most relevant articles. In paper, we propose the use of natural language processing (NLP) in the process of clustering scientific articles. Our study looks at the impact of clustering on the actual keywords provided by the authors – we analyse both the application of NLP to the abstract itself and to the introduction, followed by clustering using the K–means algorithm. As a result of experiments conducted on more than 1500 scientific articles, we have shown that our proposed approach allows articles to be approximated by their subject matter as a result of the clustering performed. The obtained results show that the best distribution of scientific articles can be obtained using the TF-IDF measure, and the worst - using TF measure.

© 2022 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 26th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2022)

Keywords: clustering, natural language processing, scientific articles

1. Introduction

The collection of scientific publications is increasing every day. The growth of the internet has also boosted the number of articles published online. This has expanded the possibility of easier access to scientific papers, while at the same time inundating the reader with too much information. This has created a need for a more precise search of the available content narrowing down the subject matter of interest. A researcher exploring a particular topic, has to look through a lot of articles on similar subjects, and it is important that he can find them efficiently and effectively. Unfortunately, it often turns out that the articles he finds may be significantly different from the subject matter. Searching for relevant articles is problematic especially for young researchers who are not yet experienced in research work and in good article filtering.

* Corresponding author.
E-mail address: barbara.probierz@ue.katowice.pl

1877-0509 © 2022 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 26th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2022)
10.1016/j.procs.2022.09.403

Another problem for modern researchers is that often only abstracts are free and access to the full article is paid. The abstract itself may be too general and not sufficiently develop the specific topic. It is only after reading the full article that the researcher can determine if the text is of value to them and contains the information they need. This process involves slowing down research and increasing the time it takes to write a scientific paper based on the information found. Additionally, it may not be enough to define the subject of articles only on the basis of the keywords declared by the authors. This is often a problem faced by editors in the process of adjusting an article to the competences of reviewers. Declared words may turn out to be too general for the recipient, and one word may have many meanings. That is why it is so important to group articles properly at the beginning of the editorial process.

Our goal is to automatically cluster scientific articles according to subject matter identified from the content of the abstract or an introduction using natural language processing (NLP). In our approach, we analyse the distribution of articles with comparison to real keywords when clustering into three, five or seven clusters. Additionally, we perform a comparison related to the applied NLP measures using TF-IDF, TF and binary measure.

We want to achieve this goal in order to improve the assignment of new articles to the correct editors in the journal in the future. In addition, to make it easier for readers to find similar articles not only on the basis of the authors' own declarations, but by automatically analysing the content of the articles.

This article is organized as follows. The 1 section introduces the problem considered in the paper. The 2 section provides an overview of related scientific article clustering. In the 3 section, we present NLP methods as data preprocessing and word weighting measures. The section 4 describes algorithm to clustering. In the section 5, we explain our methodology based on the use of NLP to analyze a text document and the use of machine learning algorithms for clustering. In the 6 section, we present and discuss the experimental results. Finally, in the section 7, we conclude with general remarks on this work, and a few directions for future research are pointed out.

2. Related works

Clustering of scientific publications is a problem that has received a lot of attention in the bibliometric literature. The main goal is to create clusters containing a set of publications that are thematically related to each other. Publications can be grouped based on occurring words in the titles, abstracts or full texts [12, 40], based on citation relations [42], and sometimes even based on a combination of different types of relationships [32, 43].

The authors [21] noted that the feature selection method based on frequency statistics has limitations in improving the clustering algorithm. Therefore, they proposed a term weighting method based on keyword features, where documents containing the same keywords are grouped into candidate clusters, and then a new cluster is created by removing irrelevant documents. The [15] presents an SVM algorithm that trains a clustering algorithm by adjusting the similarity measure of pairs of elements. The authors [19] combined classification and clustering methods to develop a method to classify scientific articles based on cluster topics. The paper [37] proposed an application for article clustering, where new keywords were selected from those that appear most frequently in an article. Based on these words, the cosine similarity to the keywords selected by the authors of the analyzed articles was calculated.

Researchers [28] noted that the traditional keyword-based approach to document grouping can produce unstable and imprecise results. Therefore, they proposed a technique of grouping short texts, which is useful for grouping abstracts of scientific articles. The obtained results showed that abstracts can be grouped with almost as good results as full texts, and the application of the K-means method gives better results than the method of k nearest neighbors. The authors [14] have proposed a new approach to clustering of short texts that is based on the ant trees. It integrates the information about the figure ratio and the concept of cluster attractiveness at different stages of the clustering process. Moreover, in cases where clustering techniques are applied to corpora containing very short documents, additional difficulties arise due to the low frequency of term occurrences in the [13] documents. For the problem of short texts clustering, a new variant of the k-means algorithm has been proposed [3].

In the paper [38], the authors decided to combine clustering with ranking to group similar documents together and then select the best documents according to the ranking. This provided a display of ranked texts according to the cosine similarity between each pair of documents in each cluster. In order to improve the performance of document clustering, a new model consisting of a document index graph and an incremental document clustering algorithm based on maximizing the tightness of clusters by carefully observing the similarity distribution of document pairs inside the clusters was proposed in the paper [16].

Some studies [1, 2] have proposed a method of collective extraction of topics from scientific publications based on cluster optimization. Statistical measures such as TF-IDF and word frequency as well as n-gram statistical measures [11] are used for keyword extraction. Automatic topic extraction from scientific publications provides a very concise summary of cluster content, but topic extraction allows you to define the boundaries of scientific disciplines [45].

3. Natural language processing

Text preprocessing is one of the main components of many text mining algorithms. It typically consists of tasks such as tokenization, lemmatization, and stemming [8]. After extracting a sentence from a sequence of characters, the next step is the tokenization process, which results in breaking the text into parts (called tokens) in the form of single words or sequences consisting of several words. The $n - gram$ model is used for the tokenization process, where n is the number of words making up the sequence and $gram$ is a single word [44]. Depending on the grammatical structure of the language the text comes from, the normalization process [27] should be performed. The commonly used normalization methods in NLP are stemming and lemmatization. These involve changing different forms of words. This aims to simplify them and ultimately reduce them to their basic form [27].

3.1. Stemming and lematization

Stemming is the process of changing all words with the same stem to a common form, usually by removing from each word its derivational and inflectional suffixes [27]. The main purpose of stemming is to transform various word forms of a word such as noun, adjective, verb, adverb, etc. into its root form [20]. However, in automatic morphological analysis, the stem of a word may be of less interest than its suffixes, which can be used as clues to grammatical structure [27].

Lemmatization aims to obtain the lemma of a word. This process involves reducing word forms to their root form by recognizing the part of speech and context of the word in a given sentence. In fact, the algorithm that converts a word to its linguistically correct core is called a lemmatizer. Lemmatization refers to the correct performance using vocabulary and morphological analysis of words most often aiming to remove only the inflectional endings and return the lemma [20].

Statistical methods involving counting the occurrence of words in texts are commonly used in NLP. They provide the necessary variables used in string processing. There are different ways to represent documents, among which the most commonly used representation is in the form of a graph or a vector of terms. In the works [30, 31], sets of n terms are created, from which an acyclic graph is then constructed and the similarity between vertices is computed. The authors of [4] in their paper presented a text graph that contains information about the order, frequency and context of terms. On the other hand, the paper [39] compares the vector approach with the graph approach. The document representation as a term vector is composed of terms, and the values of the vector are the corresponding term weights. This representation can be shown as one of three measures of word weights

3.2. Word weighting measures

The Term Frequency (TF) measure is a simple but effective term evaluation function. It is based on counting how many times each term appears in all documents [5] and is defined by the formula:

$$TF_{i,j} = \frac{n_{i,j}}{\sum_k n_{k,j}}, \tag{1}$$

where $n_{i,j}$ is the raw count of a term t_i in the document d_j and the denominator is the sum of the raw count of all $n_{k,j}$ terms in the document d_j. The larger the number, the more relevant the term is - this measure allows us to accurately determine the degree to which a term belongs to a document.

Another measure is the Inverse Document Frequency (IDF) which is a binary representation. It is based on the number of processed n_d documents and the number of documents with at least one term $\{d : t_i \in d\}$. It is one of the most useful and widely used concepts in information retrieval [5, 29]. It is expressed by the formula:

$$IDF_i = \log \frac{n_d}{\{d:t_i \in d\}} \tag{2}$$

The attribute takes a value of 1 if the term is present in the document, and a value of 0, if it is not.

The third measure is Term Frequency-Inverse Document Frequency (TF-IDF) which is one of the most widely used term weighting schemes in modern information retrieval systems. Despite its popularity, TF-IDF is often considered an empirical method, especially from a probabilistic point of view with many possible variations [6]. By definition, TF-IDF is a metric that multiplies two measures TF and IDF. TF-IDF captures relevance between words, text documents, and individual categories [46]. TF-IDF is expressed by the formula:

$$(TF - IDF)_{i,j} = TF_{i,j} \times IDF_i \tag{3}$$

The frequency of $TF_{i,j}$ terms is multiplied by the inverse of the frequency in the text IDF_i. This measure also determines the term weight locally. This is to determine lower weights for words that appear multiple times in a single document compared to weights for words that appear in multiple documents.

4. Clustering

Clustering is an unsupervised learning technique in which labels are unknown and in some cases do not even exist [7]. It deals with clustering objects that are similar to each other and at the same time dissimilar to objects belonging to other clusters. A cluster is usually identified by a centroid, which is the center of the cluster [24]. Data clustering is a difficult problem in unsupervised pattern recognition because clusters can have different sizes [17]. In addition, the number of clusters often needs to be determined before running the algorithms. Clustering is the process of identifying natural groups or clusters based on some measure of similarity [35]. Therefore, similarity measures are the basic elements of most clustering algorithms [18].

Many clustering algorithms have been developed, and most of them are based on two techniques called hierarchical clustering and partitioned clustering. The first technique involves splitting a large cluster (containing all objects) into smaller clusters until a set of single-element clusters is obtained. The merge technique, on the other hand, is the inverse of the split technique. Its construction starts with the elements of the set, which are stored on leaves and interpreted as singleton sets. Then there is a step during which two similar subsets that are stored at the nodes are merged. This process is repeated until the root of the tree is reached, which contains all the elements of the set [34]. Note that hierarchical clustering algorithms do not require a predetermined number of clusters.

Partitioned clustering algorithms devide the dataset into a number of clusters according to a selected criterion, called fitness measure, that directly affects the nature of clustering [33]. These can be treated as optimization problems because they attempt to minimize the selected criteria. Unfortunately for optimization problems, they are mostly NP-hard and combinatorial [25]. One of the partitioning algorithms is the K-means method. This method is a widely used clustering technique that aims to minimize the mean square distance between points in the same cluster [9]. This problem is NP-hard, and its solution was proposed by Lloyd in [26] by using local search. The K-means algorithm clusters the data by attempting to separate the samples into n groups of equal variance, minimizing a criterion known as inertia or intra-cluster sum of squares:

$$\sum_{i=0}^{n} \min_{\mu_j \in C}(\|x_i - \mu_j\|^2) \tag{4}$$

where the K-means algorithm divides a set of n samples x into disjoint clusters C, each of which is described by the average μ of the samples in the cluster. The averages are commonly referred to as cluster centroids. The operation of K-means involves a loop between two steps - in the first step, each sample is assigned to its closest centroid, and the second step creates new centroids by taking the average value of all samples assigned to each previous centroid. The difference between the old and new centroids is then calculated, and the algorithm repeats these last two steps until the value is less than a threshold. The work [9] extends the K-means method with a simple randomized seeding technique. It solves the problem of possible convergence to a local minimum by initializing the centroids so that they are far apart.

The K-means algorithm is one of the unsupervised learning algorithms that, due to its efficiency and simplicity, is often chosen for grouping text documents [19, 23]. The research [10] showed that the K-means algorithm gives better results for document clustering than some other known clustering algorithms, e.g. K-medoids. The authors [22] created a system that includes Term Frequency-Inverse Document Frequency (TF-IDF) and Latent Dirichlet allocation (LDA) schemas to calculate the importance of each article, and research articles on similar topics are grouped using

the K-means algorithm. The experimental results showed that the proposed system can classify the papers with similar subjects according to the keywords extracted from the abstracts of papers. In our other research [36], we have shown that the use of the K-means algorithm allows, by analyzing text from the content of books, to effectively group books on a similar subject.

5. Proposed approach

The proposed approach consists in automatic clustering of scientific articles based on the conducted text analysis. For this purpose, we want to use NLP techniques and the K-means algorithm as a machine learning algorithm for document clustering. The first step in the proposed approach is to convert all scientific articles collected as PDF files to text files. Then, the content of the articles is divided into three collections. The first set contains only the abstract part of each article, the second set is the introduction section of the articles, while the third set is the full content of the articles, but without the abstract. Additionally, keywords are extracted from the articles that will be used to evaluate the clustering work.

For all three datasets, data preprocessing is performed using the NLP methods described in 3. First, all texts are initially cleaned, by making all characters the same size, removing numbers, punctuation and other non-letter characters. Then the text is tokenized and the tokens are normalized. In our approach, a token is a single word that is then normalized using the stemming and lemmatization algorithms (described in 3.1). Additionally, the most popular natural language words are deleted according to the StopWords list, such as *per*, *well*, *whereas*, *yet*, *s*, *every*, *as*, *of*, *is* etc.

Only on such prepared data, we can perform the experiments described in the section 6. To create a matrix of token counts from data sets, we use the binary, TF and TF-IDF measures (described in 3.2). The obtained matrices will be the basis for creating clusters by applying the K-means algorithm (described in 4). In our approach, we analyze the distribution of articles depending on the specified number of clusters. It is important due to the fact that in this paper we check whether the number of clusters has an impact on bringing the articles closer to their subject as a result of the clustering. In order to evaluate the proposed approach, we conduct additional research as clustering on real keywords taken from scientific articles.

To evaluate the clustering work we propose to create a connection matrix between articles. In this case, we use the actual keywords from the articlesum as their thematic intersection, then we check if the subsequent articles share some of the keywords. The result is a connection matrix, an example of which is presented in Fig. 1 (a). In the columns and rows, there are consecutive articles, for example described as $a1$, $a2$, $a3$, $a4$ and $a5$. On the other hand, the intersections of columns and rows contain information about the number of real keywords, which are the same for both articles. The diagonal is not calculated because it is an intersection of the same article, so all keywords would match. In our example, for articles $a1$ and $a3$ there are 3 same, real, keywords; for $a2$ and $a4$ there is one same keyword, and for $a3$ and $a4$ there are no common keywords; etc.

Fig. 1 (b) presents an example visualisation of connections between articles based on their keywords. Articles (labelled, as $a1$, $a2$ etc.) have keywords (labelled, as k). Some keywords are found in two or more articles – this situation causes a connection between articles. In this case, Fig. 1 (b) corresponds exactly to the connection matrix shown in Fig. 1 (a), where e.g. $a1$, $a4$ and $a5$ have the same common keyword, and $a1$ and $a3$ have three identical keywords.

To assess the level of connections between articles, we propose to determine the ratio of the number of connections (connection coefficient). For the connection matrix shown in Fig. 1 (a), there are 10 possible connections ($\frac{n^2-n}{2}$, where n is the number of articles), this is half the number of cells of the matrix without diagonal. By calculating the number of cells in which there is a number greater than 0 (still, off the diagonal), we obtain the number of actual connections – in the example shown it is 6 cells. On this basis, it is possible to calculate the connection coefficient, i.e. the ratio of the number of actual connections to the number of all possible connections.

The number of all connections can be determined from equation (6), the number of actual connections from equation (5). Therefore, the connection coefficient is written as (8).

$$all = \frac{n^2-n}{2}, \tag{5}$$

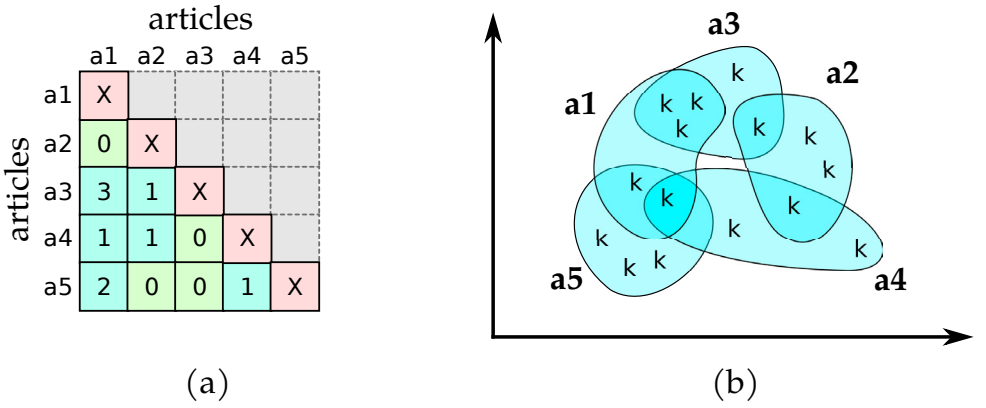


Fig. 1. Example for determining connections between articles (a) connection matrix; (b) visualisation of connections between articles based on their keywords. keywords.

where n is the number of articles.

$$actual = \sum_{c}^{all} cell(c), \tag{6}$$

where c is the next cell of the matrix and $cell(c)$ returns a number according to the equation (7).

$$cell(c) = \begin{cases} 1 \text{ , if } c > 0 \\ 0 \text{ , otherwise} \end{cases} \tag{7}$$

$$coefficient = \frac{actual}{all} \tag{8}$$

6. Experiments

In order to verify that our proposed approach (described in Section 5) achieves its goal, experiments were conducted on a real dataset we prepared. Experiments were conducted to evaluate how our approach is affected by the analysis of the abstract itself, the introduction section, the use of an appropriate measure (TF-IDF, TF or binary) in NLP, and the number of clusters.

6.1. Experimental design and data set

For the experiments, we prepared a dataset based on 1557 real scientific articles published between 2017–2022 as open access in a journal with an impact factor above 2.0. The articles were available in PDF format, so it was necessary to perform preprocessing beforehand, processing the articles into text and extracting the text of the relevant sections. This resulted in a dataset whose parameters are shown in Table 1.

We conducted experiments for 3, 5 and 7 clusters respectively for each of the mentioned measures known from NLP and abstract and introduction. Determining the optimal number of clusters is always the difficult part, especially for a dataset with little prior knowledge [41]. In our research, we chose the number of clusters to show the differences in the obtained results for the increasing number of clusters every 2 clusters. However, finding the optimal number of clusters requires extensive research. Each time using the K–means algorithm as an unsupervised learning algorithm for a predefined number of groups, an algorithm already used, among others, to cluster books based on their content [36]. This gives a total of 18 different variants of experiments, which are described in this section. The clustering results were then evaluated based on the measure described in Section 5, i.e. connections between articles located in the same clusters. The exact results are described in Section 6.2.

Table 1. Parameters of the prepared dataset.

Feature	Value
Number of articles	1557
Number of real keywords	4921
Average abstract length (number of words):	179
Average length of introduction section (number of words):	915
Average length of article content (number of words):	8194

6.2. Results of the computational experiments

Experimental results should be evaluated in terms of the number of clusters and TF–IDF, TF and binary measures. First of all, we focus on the connection coefficient (see Eq. (8)). Tab. 2 shows the results for different measures, but each time 3 clusters for content analysis of abstracts. Note that for the 1557 articles analysed, the number of all possible connections is *all* = 1211346, then in clusters with fewer articles this number decreases significantly.

Table 2. Experimental results when divided into 3 clusters – in the case of abstracts.

	all articles	binary, cluster: I	binary, cluster: II	binary, cluster: III	TF, cluster: I	TF, cluster: II	TF, cluster: III	TF-IDF, cluster: I	TF-IDF, cluster: II	TF-IDF, cluster: III
# articles	1557	130	689	738	1	1554	2	796	283	478
# keywords	4921	463	2309	2436	4	4913	6	2647	1014	1562
max con.	1211346	8385	237016	271953	0	1206681	1	316410	39903	114003
actual con.	10675	198	2434	3117	0	10557	1	3497	356	1842
con. coeff.	0.0088	0.0236	0.0103	0.0114	0.0000	0.0087	1.0000	0.0111	0.0089	0.0162
con. coeff. – weighted avg.		0.0119			0.0087			**0.0123**		

Abbreviations: # articles – number of articles; # keywords – number of keywords; max con. – Maximum number of connections; actual con. – Actual number of connections; con. coeff. – Connection coefficient.

It can be noticed already at this stage that the best results are obtained for the TF–IDF measure. This is due to the fact that in the case of the binary measure we only know whether a given word was in the content of the document, while in the case of TF we know the number of occurrences of a given word (this, however, leads to large disproportions). Even greater differences in the results appear when analysing the introduction to the article. A larger number of words to be analysed results in the fact that the created clusters contain only a few documents each, and one of the clusters concentrates all the others. As can be seen, for the TF measure such a situation is already the case in the analysis of the abstract content. A cluster is formed with one document, with two and a third with all the others. Better results appear for the binary measure (although for the introduction it is already characterised by a similar behaviour as TF in this case).

In the case of our experiments, the aim was to check whether the documents in the groups converge to each other, to assess this we proposed a connection coefficient (see Eq. (8)). For the whole documents the connection coefficient is less than 1%, but after clustering the documents processed with NLP based on the binary measure we obtain connection coefficients for the groups, respectively: 2.4%, 1.0% i 1.1%. A similar situation is observed when TF–IDF is applied, but here the groups are divided into more equal groups.

By calculating the weighted average connection coefficient it can be seen that, compared to 0.88% for data without culling, we can obtain 1.19% when using the binary measure; 0.87% for the TF measure and 1.23% when using the TF–IDF measure. The values do not seem high, but with such a large connection matrix we cannot expect a high keyword coverage, so we need to evaluate the improvement of the connection coefficient, and this when using K-means culling with 3 clusters and NLP with the TF–IDF measure increases by half.

Tab. 3 and 4 present the results related to the application of the TF–IDF measure with a higher number of clusters: 5 (Tab. 3) and 7 (Tab. 4), respectively. The results presented are related to both the content analysis of the abstracts and the introduction. The weighted average connection coefficient is the largest from the point of view of previous

Table 3. Experimental results for TF-IDF measure and division into 5 clusters.

	all articles	TF-IDF, cluster: I	II	III	IV	V
Analysis of the content of the abstract						
# articles	1557	596	77	160	568	156
# keywords	4921	2065	281	582	1901	481
max con.	1211346	177310	2926	12720	161028	12090
actual con.	10675	2379	118	195	1878	246
con. coeff.	0.0088	0.0134	0.0403	0.0153	0.0117	0.0203
con. coeff. – weighted average		0.0150				
Analysis of the content of the introduction						
# articles	1557	202	729	362	174	90
# keywords	4921	737	2402	1250	542	333
max con.	1211346	20301	265356	65341	15051	4005
actual con.	10675	262	2973	1092	329	128
con. coeff.	0.0088	0.0129	0.0112	0.0167	0.0219	0.0320
con. coeff. – weighted average		0.0151				

Abbreviations: # articles – number of articles; # keywords – number of keywords; max con. – Maximum number of connections; actual con. – Actual number of connections; con. coeff. – Connection coefficient.

observations. It can be seen that the number of articles with connections is slightly higher when analyzing the whole introduction – this is particularly noticeable when clustering into a larger number of classes. On the other hand, for both introduction and abstracts, the connection coefficient increases with the number of clusters.

If we used 5 clusters the connection coefficient increases from 0.88% (before clustering) to 1.5%, i.e. almost twice as much. However, when we used 7 clusters the connection coefficient is already above 1.7%, i.e. scientific articles in clusters are connected to each other twice as often as before clustering. It should be noted that increasing the number of clusters affects the size of the clusters. This influences the density of similar articles, and thus increases the connection coefficient, but at the same time generalizes the analyzed articles to a lesser extent. This is a desirable feature e.g. in the case of many editors and diversification of their scientific interests.

7. Conclusions and future works

The aim of this study was to automatically cluster scientific articles according to the topics identified on the basis of the content of the abstract or introduction. NLP techniques were used to analyze the text, in particular measures based on word weight, i.e. TF, TF-IDF and binary. In our research, we analyzed a collection of over 1,550 scientific articles published in 2017-2022 from one scientific journal.

By applying the clustering algorithm, we obtained the distribution of articles into three, five and seven clusters and compared the results depending on the applied NLP measures. We conclude from the obtained results that the best distribution of articles can be obtained after applying the TF-IDF measure, and the worst for the TF measure. In the case of clustering into three clusters and applying the TF measure, we obtained two clusters with few articles and a third cluster that contained all other articles. We obtained better results using the binary measure, however, in later studies it showed a similar behavior to TF. Therefore, the results for TF and binary were presented for three clusters only.

Observation of our experiments with the TF-IDF measure indicates that by increasing the number of clusters, we obtain higher weighted average connection coefficient results. For both 5 and 7 clusters, much better results were obtained by clustering based on the introduction. Our results for the TF-IDF measure and 7 clusters show that scientific articles are connected to each other twice as often in clusters as before clustering. This allows us to confirm the authors' hypothesis.

Table 4. Experimental results for TF-IDF measure and division into 7 clusters.

	all articles	TF-IDF, cluster: I	II	III	IV	V	VI	VII
Analysis of the content of the abstract								
# articles	1557	101	176	287	99	353	472	69
# keywords	4921	326	662	927	359	1297	1603	265
max con.	1211346	5050	15400	41041	4851	62128	111156	2346
actual con.	10675	108	186	1131	122	312	2055	39
con. coeff.	0.0088	0.0214	0.0121	0.0276	0.0252	0.0050	0.0185	0.0166
con. coeff. – weighted average		0.0171						
Analysis of the content of the introduction								
# articles	1557	30	113	570	595	54	67	128
# keywords	4921	121	363	1888	2074	198	238	427
max con.	1211346	435	6328	162165	176715	1431	2211	8128
actual con.	10675	9	289	1533	1665	94	94	333
con. coeff.	0.0088	0.0207	0.0457	0.0095	0.0094	0.0657	0.0425	0.0410
con. coeff. – weighted average		0.0183						

Abbreviations: # articles – number of articles; # keywords – number of keywords; max con. – Maximum number of connections; actual con. – Actual number of connections; con. coeff. – Connection coefficient.

In the future, we want to extend our research to the analysis of scientific articles, based on the set of full texts of articles, and compare the results with those obtained in this work. Additionally, we noticed that the distribution of articles in individual clusters was uneven. There have been clusters with significantly more articles than others, which we believe should be explored in the future. We would therefore like to include the criterion of equivalence in order to improve clustering. In addition, we want to extend our experiments in terms of finding the optimal number of clusters, which would certainly improve the quality of our research.

References

[1] Abasi, A.K., Khader, A.T., Al-Betar, M.A., Naim, S., Alyasseri, Z.A.A., Makhadmeh, S.N., 2021a. An ensemble topic extraction approach based on optimization clusters using hybrid multi-verse optimizer for scientific publications. Journal of Ambient Intelligence and Humanized Computing 12, 2765–2801.

[2] Abasi, A.K., Khader, A.T., Al-Betar, M.A., Naim, S., Makhadmeh, S.N., Alyasseri, Z.A.A., 2021b. A novel ensemble statistical topic extraction method for scientific publications based on optimization clustering. Multimedia Tools and Applications 80, 37–82.

[3] Abdalgader, K., 2017. Clustering short text using a centroid-based lexical clustering algorithm. IAENG International Journal of Computer Science 44.

[4] Abilhoa, W.D., De Castro, L.N., 2014. A keyword extraction method from twitter messages represented as graphs. Applied Mathematics and Computation 240, 308–325.

[5] Adeva, J.G., Atxa, J.P., Carrillo, M.U., Zengotitabengoa, E.A., 2014. Automatic text classification to support systematic reviews in medicine. Expert Systems with Applications 41, 1498–1508.

[6] Aizawa, A., 2003. An information-theoretic perspective of TF-IDF measures. Information Processing & Management 39, 45–65.

[7] Alhawarat, M., Hegazi, M., 2018. Revisiting K-means and topic modeling, a comparison study to cluster arabic documents. IEEE Access 6, 42740–42749.

[8] Allahyari, M., Pouriyeh, S., Assefi, M., Safaei, S., Trippe, E.D., Gutierrez, J.B., Kochut, K., 2017. A brief survey of text mining: Classification, clustering and extraction techniques. arXiv preprint arXiv:1707.02919 .

[9] Arthur, D., Vassilvitskii, S., 2006. K-means++: The advantages of careful seeding. Technical Report. Stanford.

[10] Balabantaray, R.C., Sarma, C., Jha, M., 2015. Document clustering using K-means and K-medoids. arXiv preprint arXiv:1502.07938 .

[11] Beliga, S., Meštrović, A., Martinčić-Ipšić, S., 2015. An overview of graph-based keyword extraction methods and approaches. Journal of information and organizational sciences 39, 1–20.

[12] Boyack, K.W., Newman, D., Duhon, R., Klavans, R., Patek, M., Biberstine, J., Schijvenaars, B., Skupin, A., Ma, N., Börner, K., 2011. Clustering more than two million biomedical publications: Comparing the accuracies of nine text-based similarity approaches. PloS one 6, e18029.

[13] Cagnina, L., Errecalde, M., Ingaramo, D., Rosso, P., 2014. An efficient particle swarm optimization approach to cluster short texts. Information Sciences 265, 36–49.

[14] Errecalde, M.L., Ingaramo, D.A., Rosso, P., 2010. A new AntTree-based algorithm for clustering short-text corpora. Journal of Computer Science & Technology 10.
[15] Finley, T., Joachims, T., 2005. Supervised clustering with support vector machines, in: Proceedings of the 22nd international conference on Machine learning, pp. 217–224.
[16] Hammouda, K.M., Kamel, M.S., 2004. Efficient phrase-based document indexing for web document clustering. IEEE Transactions on knowledge and data engineering 16, 1279–1296.
[17] Jain, A.K., Duin, R.P.W., Mao, J., 2000. Statistical pattern recognition: A review. IEEE Transactions on pattern analysis and machine intelligence 22, 4–37.
[18] Jain, A.K., Murty, M.N., Flynn, P.J., 1999. Data clustering: a review. ACM computing surveys (CSUR) 31, 264–323.
[19] Jalal, A.A., Ali, B.H., 2021. Text documents clustering using data mining techniques. International Journal of Electrical & Computer Engineering (2088-8708) 11.
[20] Jivani, A.G., et al., 2011. A comparative study of stemming algorithms. Int. J. Comp. Tech. Appl 2, 1930–1938.
[21] Kang, S.S., 2003. Keyword-based document clustering, in: Proceedings of the sixth international workshop on Information retrieval with Asian languages, pp. 132–137.
[22] Kim, S.W., Gil, J.M., 2019. Research paper classification systems based on TF-IDF and LDA schemes. Human-centric Computing and Information Sciences 9, 1–21.
[23] Lakshmi, R., Baskar, S., 2019. DIC-DOC-K-means: dissimilarity-based Initial Centroid selection for DOCument clustering using K-means for improving the effectiveness of text document clustering. Journal of Information Science 45, 818–832.
[24] Lee, C.Y., Antonsson, E., 2000. Dynamic partitional clustering using evolution strategies, in: 2000 26th Annual Conference of the IEEE Industrial Electronics Society. IECON 2000. 2000 IEEE International Conference on Industrial Electronics, Control and Instrumentation. 21st Century Technologies, IEEE. pp. 2716–2721.
[25] Leung, Y., Zhang, J.S., Xu, Z.B., 2000. Clustering by scale-space filtering. IEEE Transactions on pattern analysis and machine intelligence 22, 1396–1410.
[26] Lloyd, S., 1982. Least squares quantization in PCM. IEEE transactions on information theory 28, 129–137.
[27] Lovins, J.B., 1968. Development of a stemming algorithm. Mech. Transl. Comput. Linguistics 11, 22–31.
[28] Makagonov, P., Alexandrov, M., Gelbukh, A., 2004. Clustering abstracts instead of full texts, in: International Conference on Text, Speech and Dialogue, Springer. pp. 129–135.
[29] Metzler, D., 2008. Generalized inverse document frequency, in: Proceedings of the 17th ACM conference on Information and knowledge management, pp. 399–408.
[30] Mosa, M.A., Hamouda, A., Marei, M., 2017a. Ant colony heuristic for user-contributed comments summarization. Knowledge-Based Systems 118, 105–114.
[31] Mosa, M.A., Hamouda, A., Marei, M., 2017b. Graph coloring and ACO based summarization for social networks. Expert Systems with Applications 74, 115–126.
[32] Most, F., Conejo, F.J., Cunningham, L.F., 2018. Bridging past and present entrepreneurial marketing research: A co-citation and bibliographic coupling analysis. Journal of Research in Marketing and Entrepreneurship .
[33] Nanda, S.J., Panda, G., 2014. A survey on nature inspired metaheuristic algorithms for partitional clustering. Swarm and Evolutionary computation 16, 1–18.
[34] Nielsen, F., 2016. Introduction to HPC with MPI for Data Science. Springer.
[35] Omran, M.G., Engelbrecht, A.P., Salman, A., 2007. An overview of clustering methods. Intelligent Data Analysis 11, 583–605.
[36] Probierz, B., Kozak, J., Hrabia, A., 2022. A comparative study of classification and clustering methods from text of books. 14th Asian Conference on Intelligent Information and Database Systems, ACIIDS 2022, Ho Chi Minh City, Vietnam, November 28-30, 2022 , in print.
[37] Rinartha, K., Kartika, L.G.S., 2019. Scientific article clustering using string similarity concept, in: 2019 1st International Conference on Cybernetics and Intelligent System (ICORIS), IEEE. pp. 13–17.
[38] Roul, R.K., Devanand, O.R., Sahay, S.K., 2014. Web document clustering and ranking using TF-IDF based apriori approach. arXiv preprint arXiv:1406.5617 .
[39] Rumagit, R.Y., Setiyawati, N., Bangkalang, D.H., 2019. Comparison of graph-based and term weighting method for automatic summarization of online news. Procedia Computer Science 157, 663–672.
[40] Salloum, S.A., Al-Emran, M., Monem, A.A., Shaalan, K., 2018. Using text mining techniques for extracting information from research articles, in: Intelligent natural language processing: Trends and Applications. Springer, pp. 373–397.
[41] Shi, C., Wei, B., Wei, S., Wang, W., Liu, H., Liu, J., 2021. A quantitative discriminant method of elbow point for the optimal number of clusters in clustering algorithm. EURASIP Journal on Wireless Communications and Networking 2021, 1–16.
[42] Šubelj, L., Van Eck, N.J., Waltman, L., 2016. Clustering scientific publications based on citation relations: A systematic comparison of different methods. PloS one 11, e0154404.
[43] Waltman, L., Van Eck, N.J., Noyons, E.C., 2010. A unified approach to mapping and clustering of bibliometric networks. Journal of informetrics 4, 629–635.
[44] Wang, K., Thrasher, C., Viegas, E., Li, X., Hsu, B.j.P., 2010. An overview of microsoft web N-gram corpus and applications, in: Proceedings of the NAACL HLT 2010 Demonstration Session, pp. 45–48.
[45] Wang, Z., Hahn, K., Kim, Y., Song, S., Seo, J.M., 2018. A news-topic recommender system based on keywords extraction. Multimedia Tools and Applications 77, 4339–4353.
[46] Yun-tao, Z., Ling, G., Yong-cheng, W., 2005. An improved TF-IDF approach for text classification. Journal of Zhejiang University-Science A 6, 49–55.